

PROJECT BIG-ES

Barangay Integrated Governance and Environmental Sustainability with Revenue Projections Using Prophet Forecasting Model

Project Title	Project BIG-ES: Barangay Integrated Governance and Environmental Sustainability with Revenue Projections Using Prophet Forecasting Model
Prepared By	Kirk Nathaniel A. Gamorot
Date / Version	July 2026 Version 1.0

Project Overview

Project BIG-ES is a centralized web-based platform developed for Barangay Bagbag, Quezon City. It combines digital governance, environmental sustainability monitoring, and analytics-driven revenue forecasting into one system. The platform was designed by BSIT major in Business Analytics capstone and supports resident verification, e-service transactions, Material Recovery Facility (MRF) logging, urban farming tracking, and predictive planning using the Prophet forecasting model.

Problem Statement

- Resident records were handled manually, making verification, retrieval, and updating slow and prone to error.
- Barangay e-services such as clearances, residency certificates, and blotter-related processes were processed through paper-based steps that delayed service delivery.
- MRF and urban farming activities were not digitally tracked, limiting visibility into sustainability operations and income generation.
- Decision-makers lacked a simple way to interpret trend data and revenue projections for planning purposes.

Proposed Solution

Project BIG-ES addresses these issues through a verification-first workflow, role-based access control, and an integrated analytics module. Verified residents can request online services, barangay staff can manage e-service and sustainability records, and administrators can view descriptive reports and Prophet-based forecasts. The platform also includes a Smart Report feature that converts charts into plain-language summaries for easier interpretation.

Key System Modules

Module	Notes
Resident Verification	To access BIG-ES, residents must first create an account where it will be validated by an admin.
E-Services	Blotter, Barangay Clearance, Certificate of Residency, Certificate of Indigency, Barangay ID, Good Moral Certificate, Toda Clearance, Liquor Permit, Public Utilities Clearance, Construction Clearance, Working Permit
MRF and Urban Farming Tracking	Waste such as Cardboard, Glass, Metal, Plastic, Paper can be digitally recorded. These records then translated to dashboards for revenue reports.

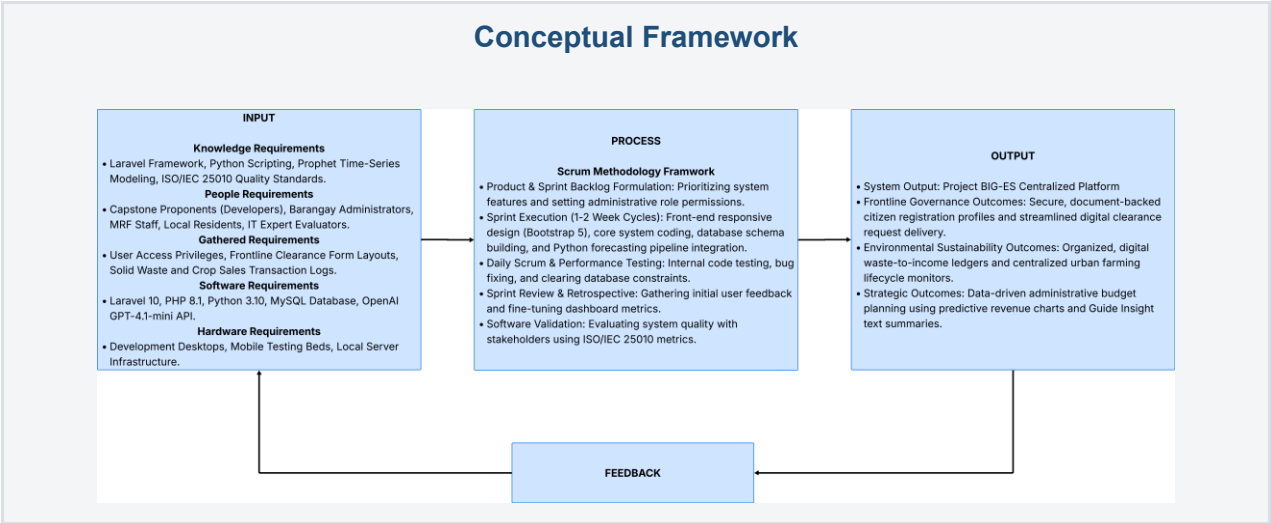
Analytics Dashboard	Charts found on the website are Trend Lines, Donut Chart, Horizontal Bar Chart, Clustered Bar Charts, Stacked Bar Chart, Area Chart, Forecast Charts.
Smart Report	Automatically translates charts, revenue trends, and forecasting results into easy-to-understand text summaries, enabling users to quickly interpret analytical insights without requiring technical or data analysis expertise.

Technology Stack

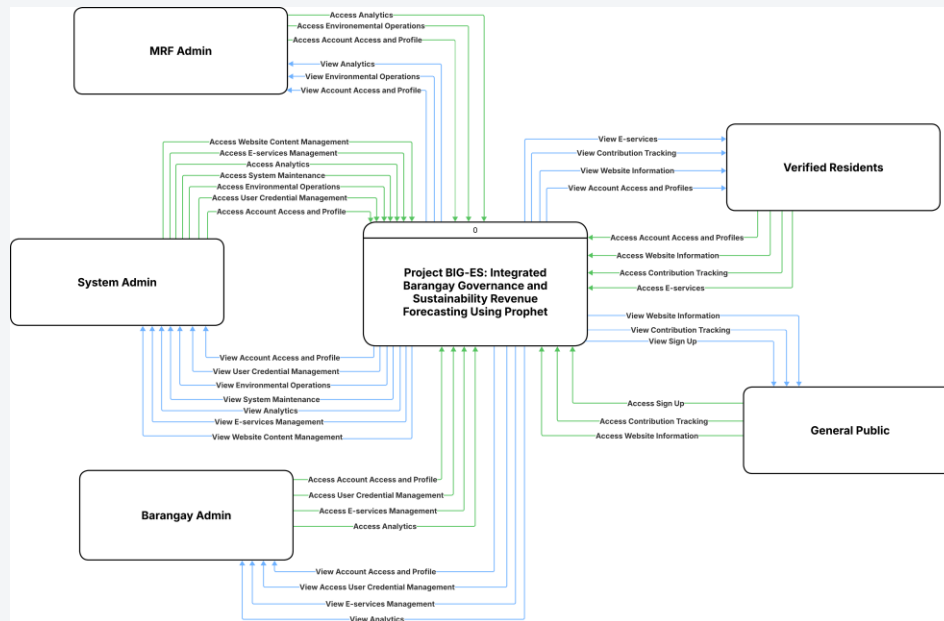
Section	Details / Placeholder
Frontend	Bootstrap
Backend / Framework	Laravel, PHP 8.1
Database	MySQL
Analytics and Forecasting	Python, Prophet
AI Support	OpenAI GPT-4.1-mini API

System Design Figures

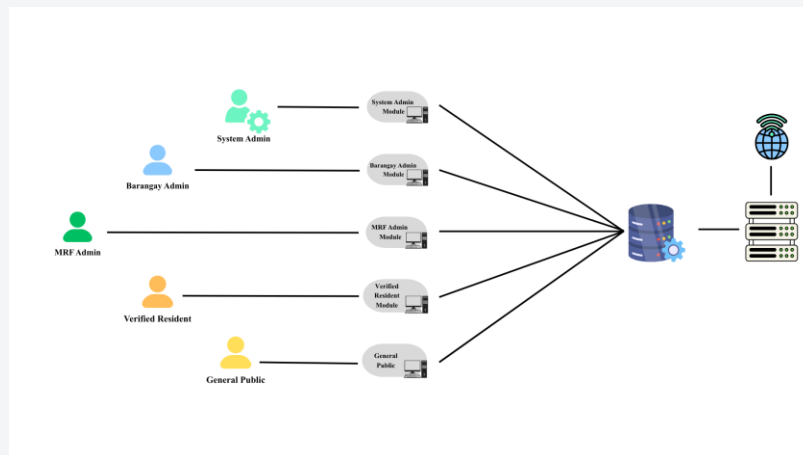
Insert the final diagrams below. Replace the placeholders with exported PNGs from your manuscript or design files.



Context Flow Diagram



System Architecture



Evaluation Snapshot

The project was evaluated with 83 respondents using the ISO/IEC 25010 software quality standard. The study reported a Grand Mean of 3.29, interpreted as “Strongly Agree.” The abstract also notes strong ratings for system flexibility (3.57), resident reliability (3.36), functional suitability (3.66), and security (3.60).

My Contributions

- Responsible for project planning, team coordination, quality assurance, research and manuscript preparation, technical documentation, and monitoring project progress from development through final defense.

- Led a four-member capstone team across four academic semesters while authoring most of Chapters 1–6 of the manuscript, designing the system evaluation, creating the majority of figures and tables, conducting quality assurance and test case development, and coordinating weekly project progress through Functional Requirements Specification (FRS) tracking.

Future Enhancements

- Cross-Platform Mobile Application – Extend the platform into a dedicated mobile application to improve accessibility and enable residents to access barangay services through smartphones.
- Multi-Barangay Deployment – Expand the system into a cloud-based multi-tenant platform capable of supporting multiple barangays under a shared digital infrastructure.
- OCR-Assisted Resident Verification – Integrate Optical Character Recognition (OCR) to automate ID and proof-of-residency validation, reducing manual verification efforts.
- Continuous System Maintenance – Perform regular security audits, database optimization, and periodic retraining of the Prophet forecasting model to maintain system performance and accuracy.
- Enhanced AI Capabilities – Upgrade from the OpenAI GPT-4.1-mini API to a higher-tier model to improve the availability and responsiveness of the Smart Report feature.
- Filipino Language Support – Introduce multilingual support through a Filipino language option to improve accessibility and encourage greater community adoption.